

Jenkins Pipeline to Pull Docker Image & Run Container

Objective

To create a Jenkins pipeline that pulls a Docker image from Docker Hub and runs it as a container.

Prerequisites

1. Jenkins installed and running 2. Docker Desktop installed and running 3. Git installed 4. Basic knowledge of Jenkins pipeline

Step 1: Verify Docker Installation

Run the following command in Jenkins or system terminal:

```
docker --version
docker ps
```

Step 2: Configure Jenkins Access to Docker

Ensure Docker is added to system PATH and Jenkins has permission to access Docker.

Step 3: Create Jenkins Pipeline Job

1. Open Jenkins Dashboard 2. Click on 'New Item' 3. Enter name: docker-pipeline 4. Select Pipeline 5. Click OK

Step 4: Add Pipeline Script

```
pipeline {
    agent any

    stages {
        stage('Check Docker') {
            steps {
                bat 'docker --version'
            }
        }

        stage('Pull Docker Image') {
            steps {
                bat 'docker pull nginx'
            }
        }

        stage('Run Container') {
            steps {
                bat 'docker run -d -p 8085:80 --name my-nginx nginx'
            }
        }

        stage('Verify Container') {
```

```
    steps {  
      bat 'docker ps'  
    }  
  }  
}
```

Step 5: Run the Pipeline

Click on 'Build Now' and observe the pipeline stages.

Step 6: Verify Output

Open browser and go to: <http://localhost:8085>

Common Errors and Fixes

1. Docker not recognized → Add Docker to PATH 2. Port already in use → Change port 3. Container already exists → Remove container using `docker rm -f`

Conclusion

Successfully implemented Jenkins pipeline to pull Docker image and run container.